

5-Fold Cross Validation Evaluation Report (Ensemble + Calibration)

Started: 2026-04-14 14:51:54

Finished: 2026-04-14 17:08:17

Model Architecture: ResNet50

Phase 1 LR: 0.01 | Phase 2 LR: 1e-06

Aggregate Performance across 5 Folds (Individual Optimized Thresholds):

Average Accuracy: 0.9203 ± 0.0067

Average Sensitivity: 0.9373 ± 0.0091

Average Specificity: 0.9027 ± 0.0188

Average AUC: 0.9711 ± 0.0022

CALIBRATED ENSEMBLE TEST METRICS:

Forced Decision Threshold: 0.5000

Accuracy : 0.8987

Precision : 0.9239

Sensitivity (Recall) : 0.8725

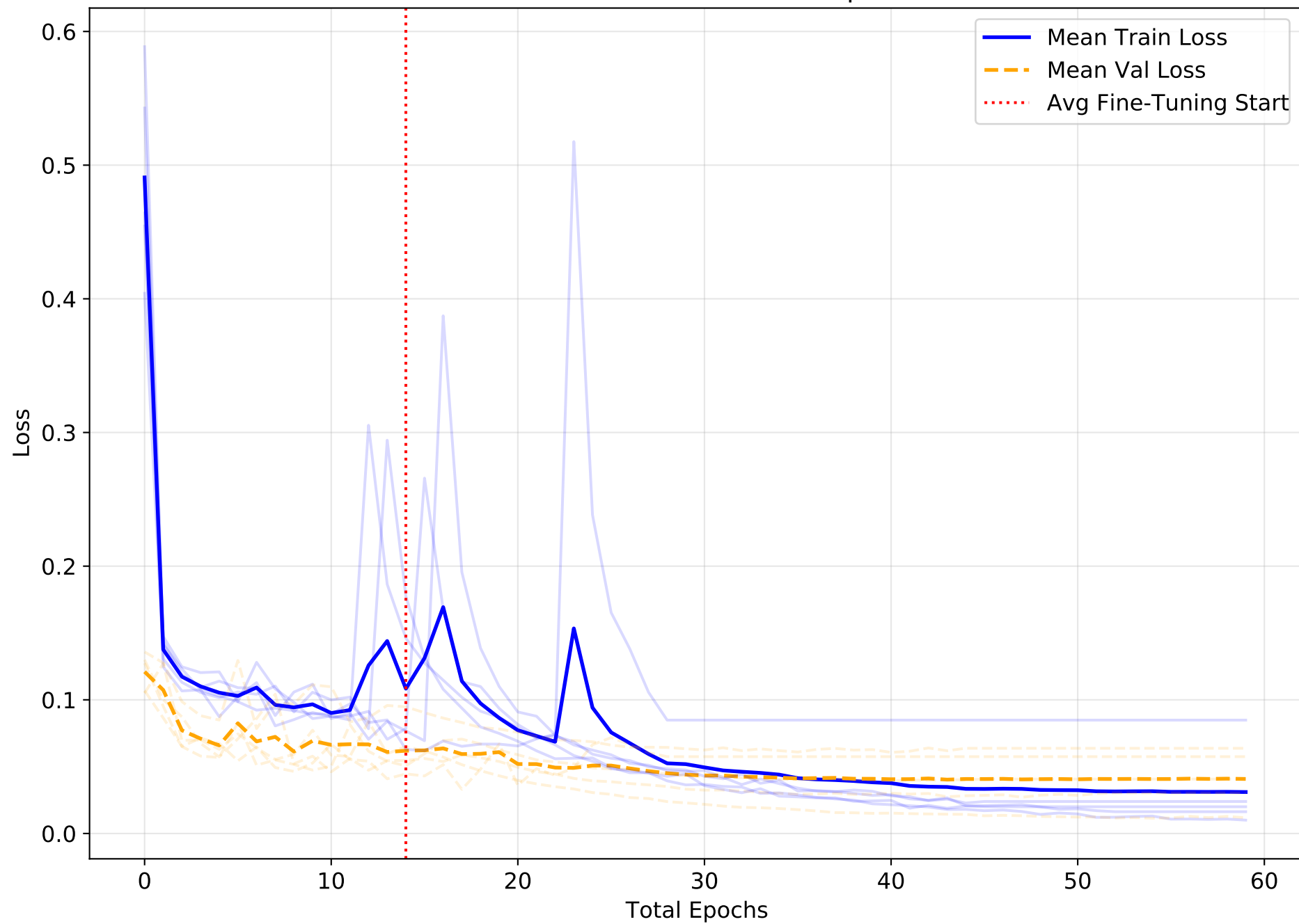
Specificity : 0.9257

Ensemble AUC : 0.9725

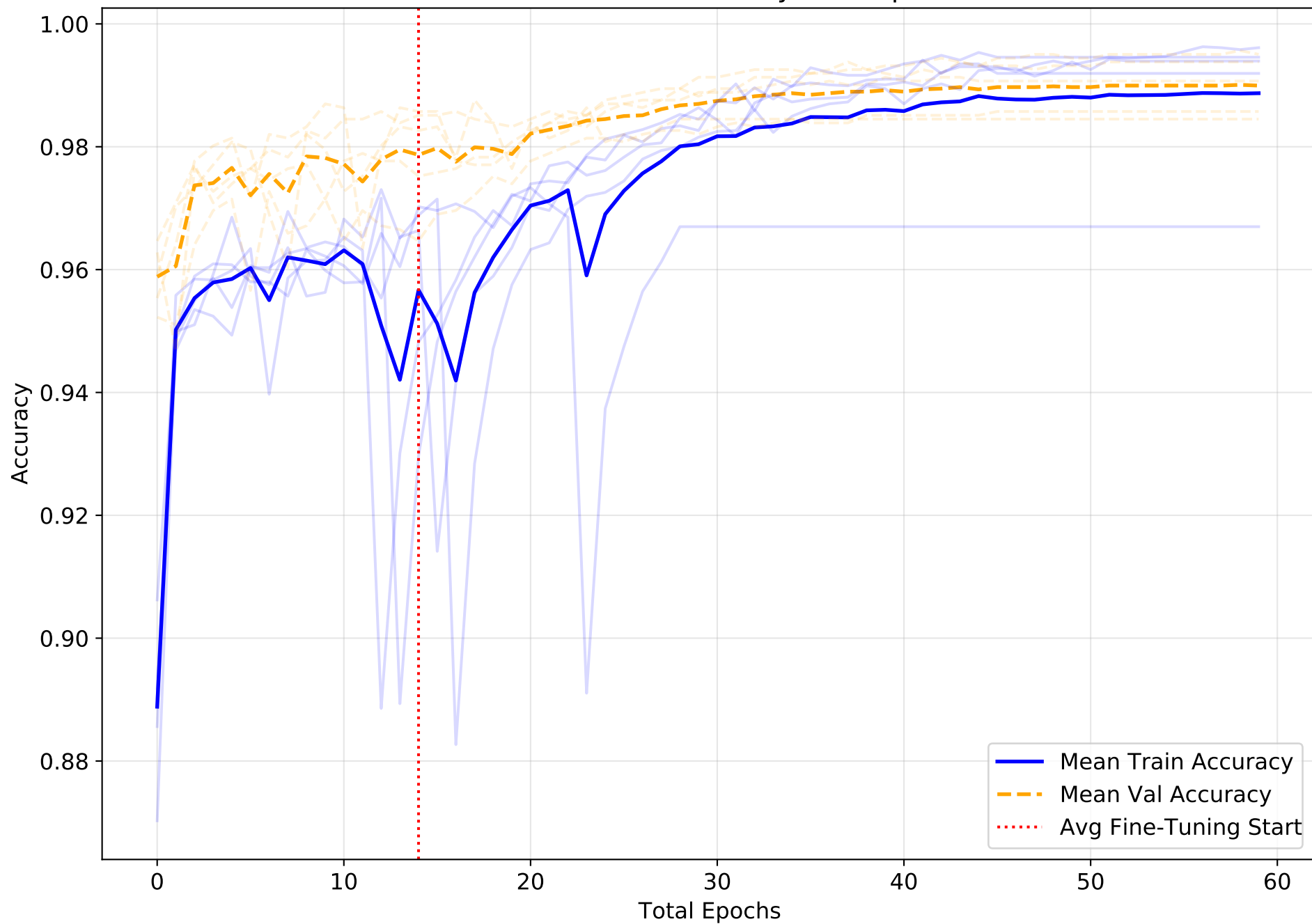
Classification Report (at threshold 0.5000):

	precision	recall	f1-score	support
Normal	0.88	0.93	0.90	296
Cataract	0.92	0.87	0.90	306
accuracy			0.90	602
macro avg	0.90	0.90	0.90	602
weighted avg	0.90	0.90	0.90	602

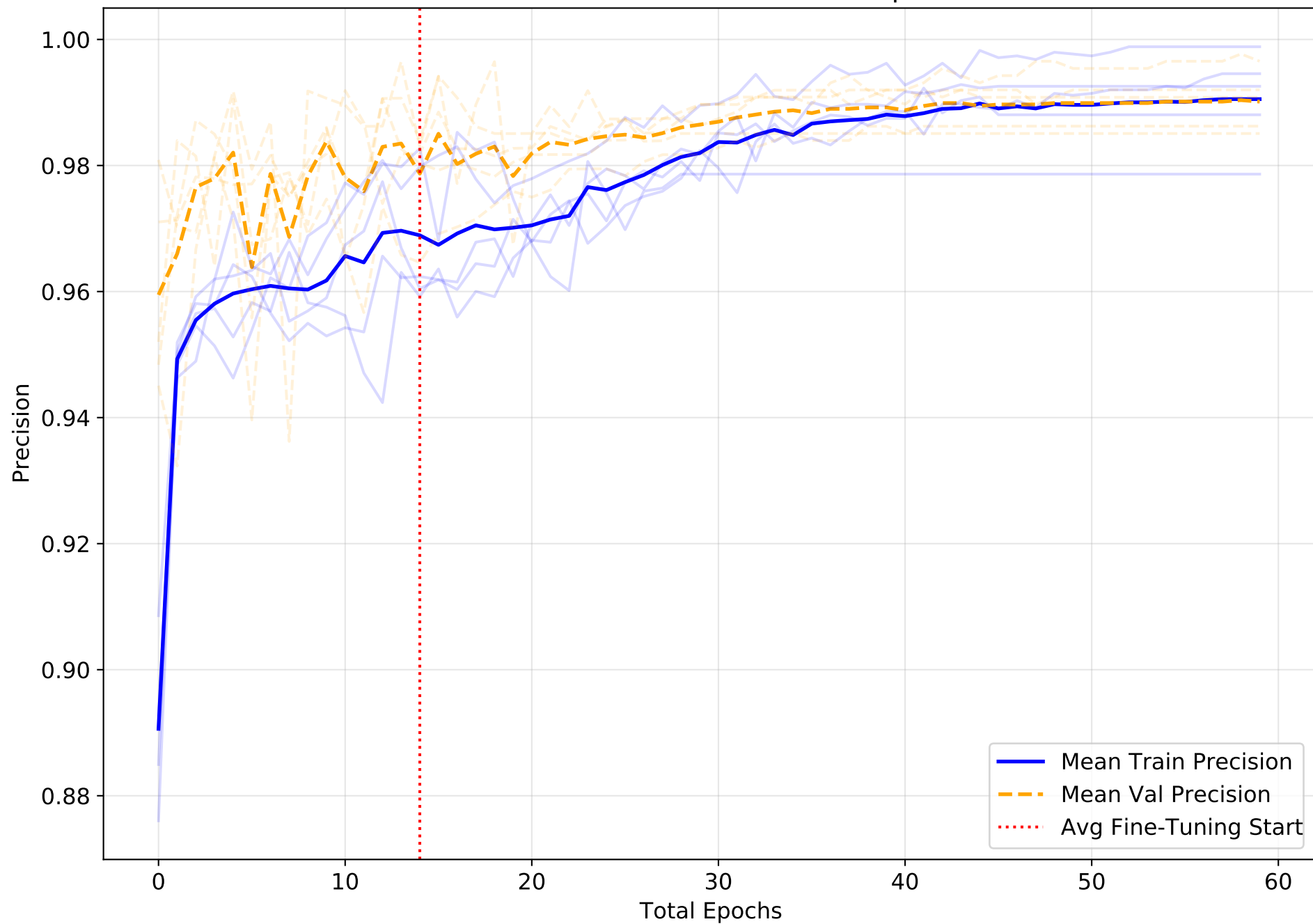
Cross-Validated Loss Over Epochs



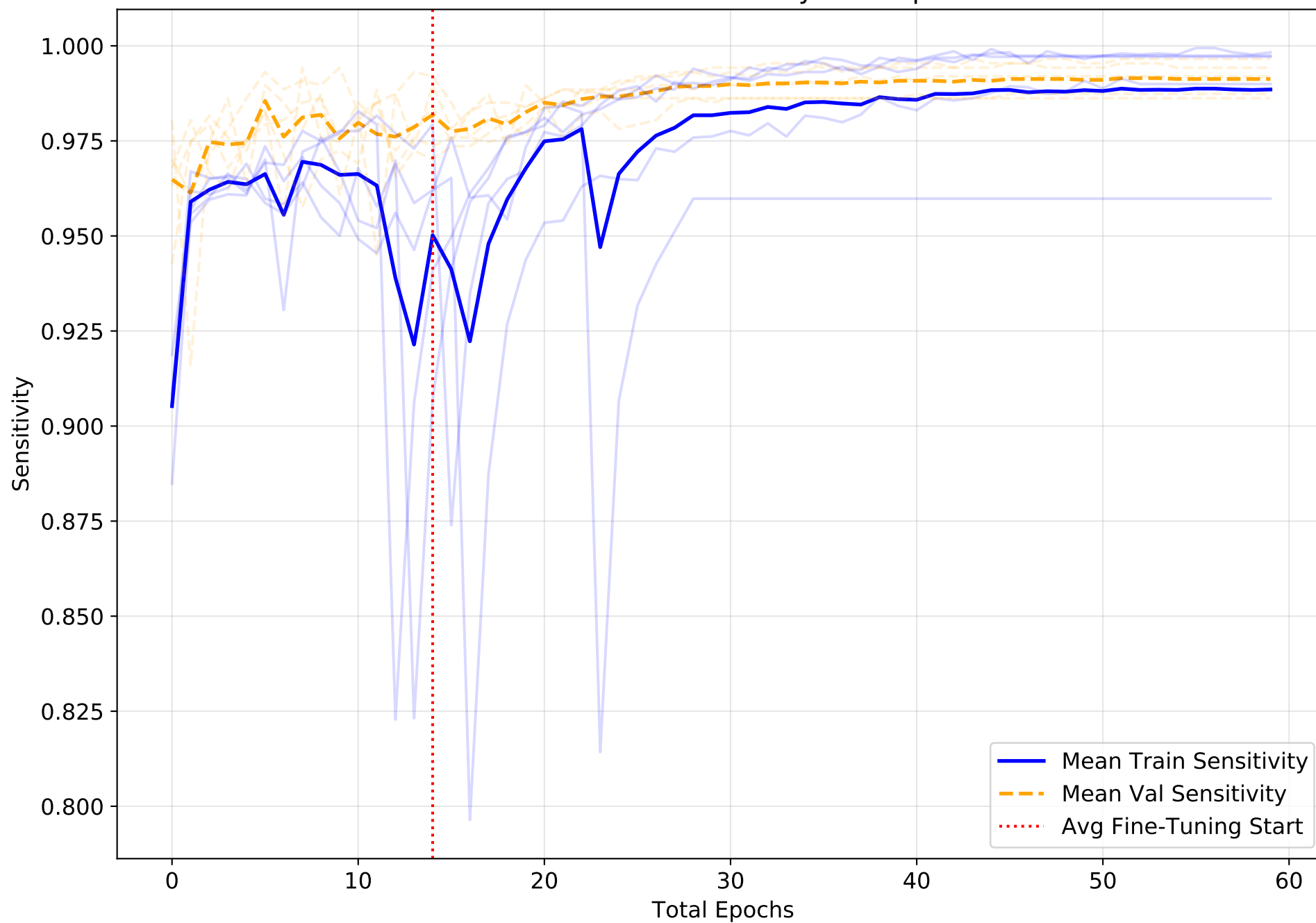
Cross-Validated Accuracy Over Epochs



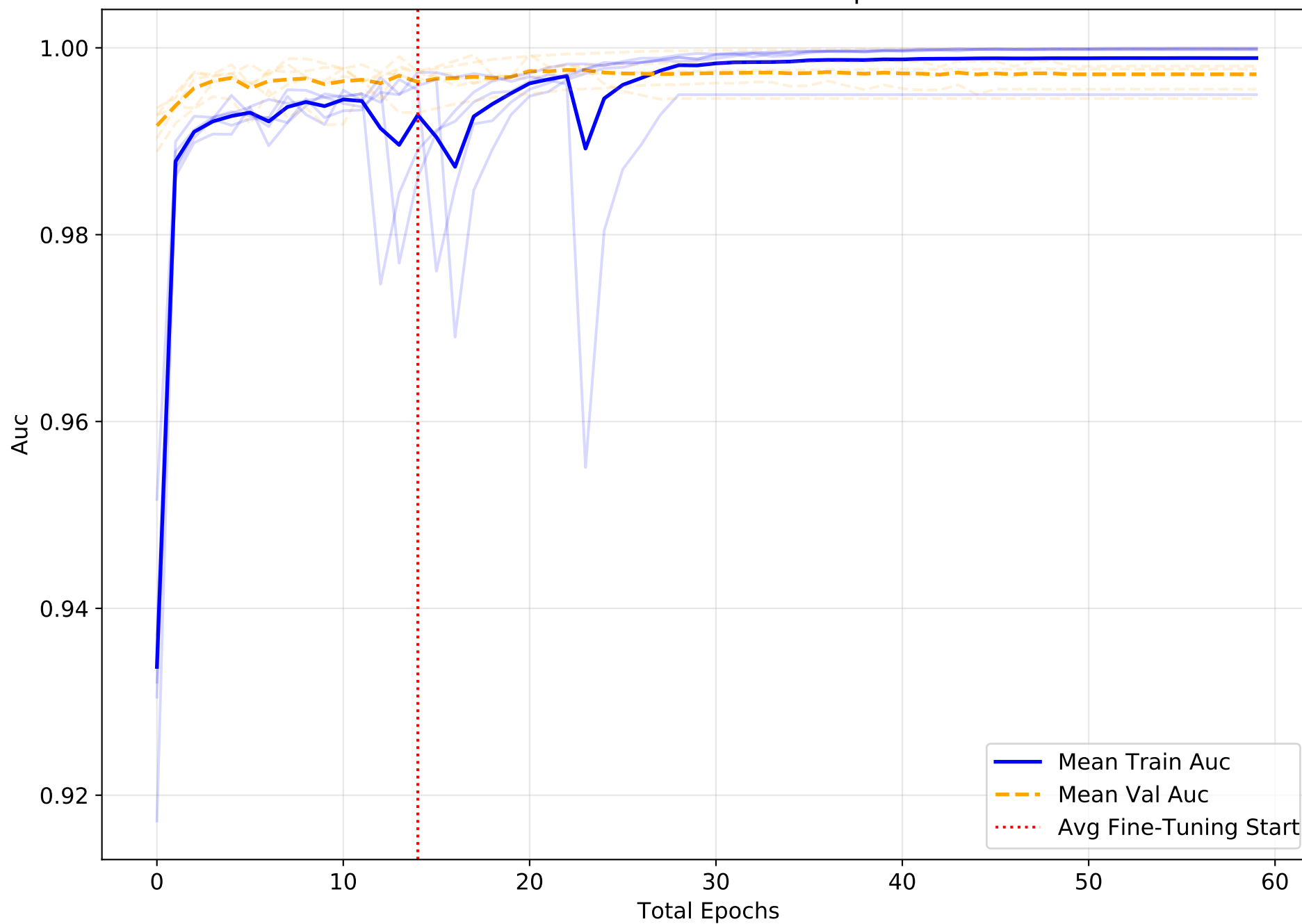
Cross-Validated Precision Over Epochs



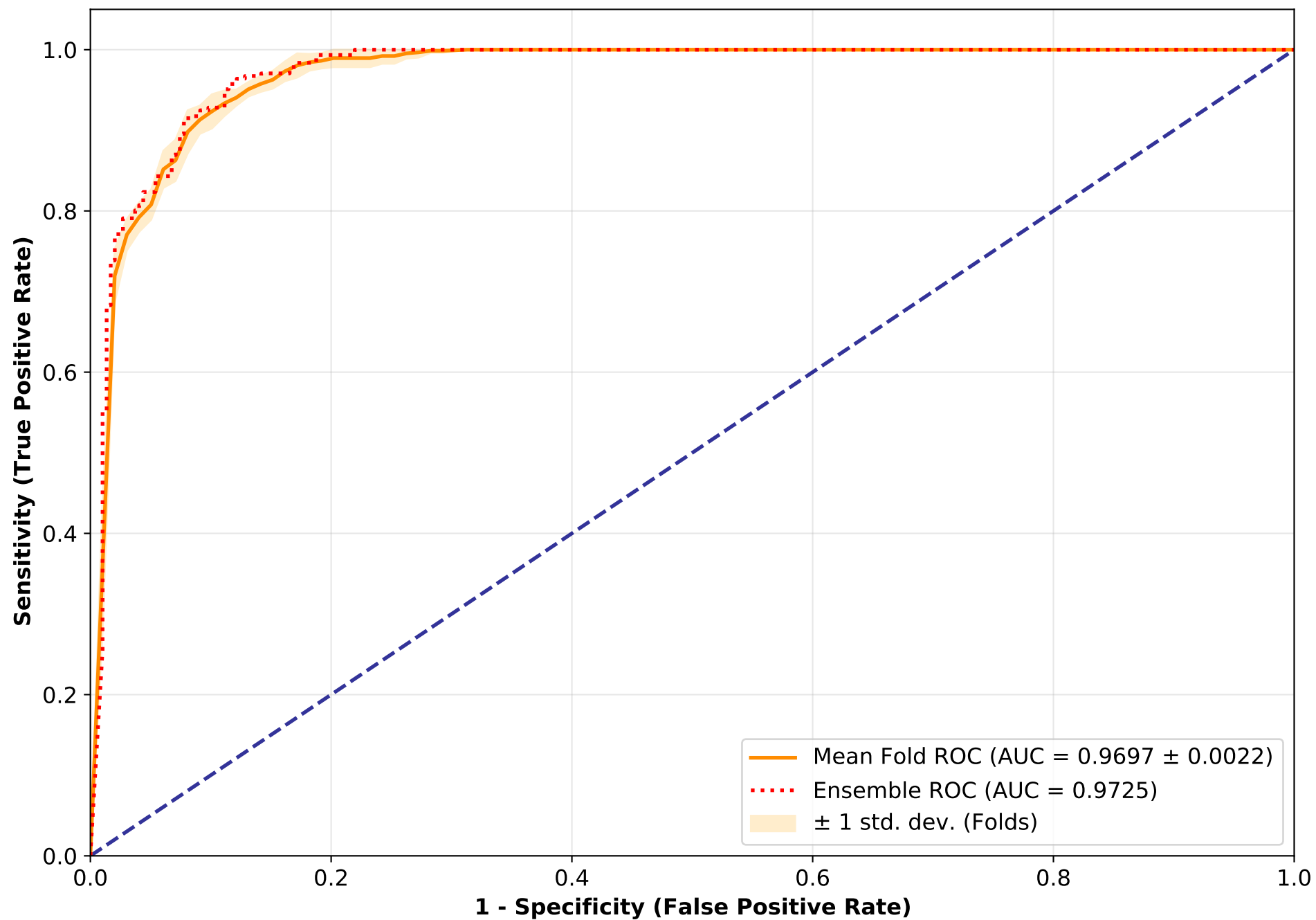
Cross-Validated Sensitivity Over Epochs



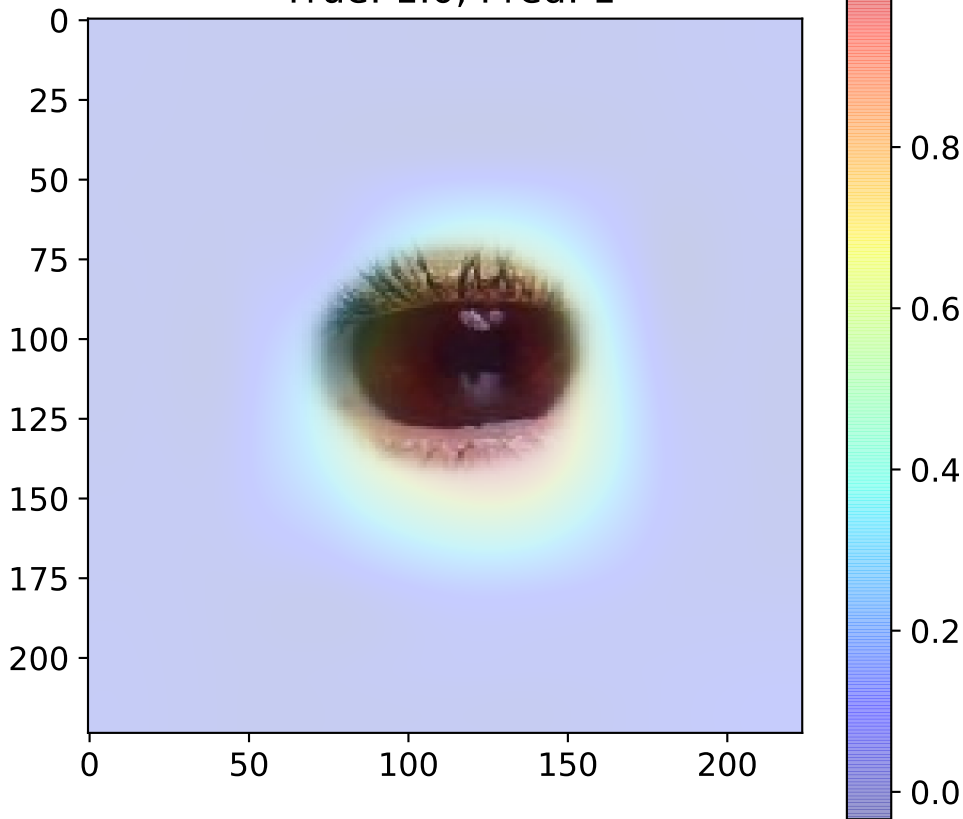
Cross-Validated Auc Over Epochs



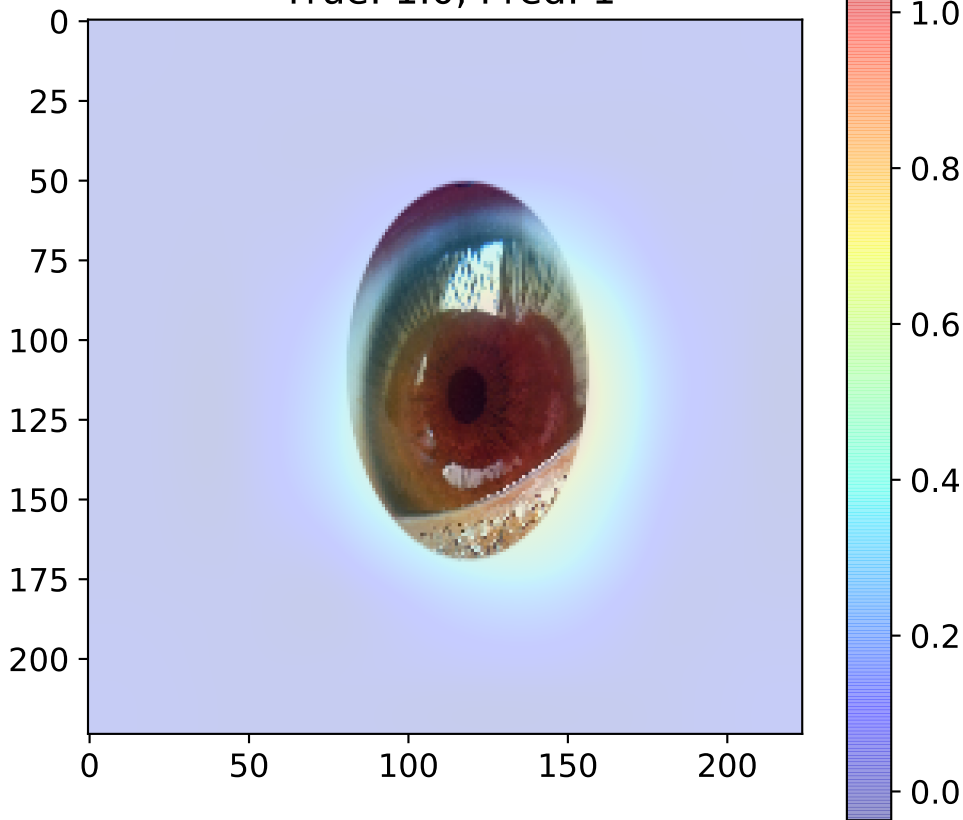
Cross-Validated ROC Curve - ResNet50



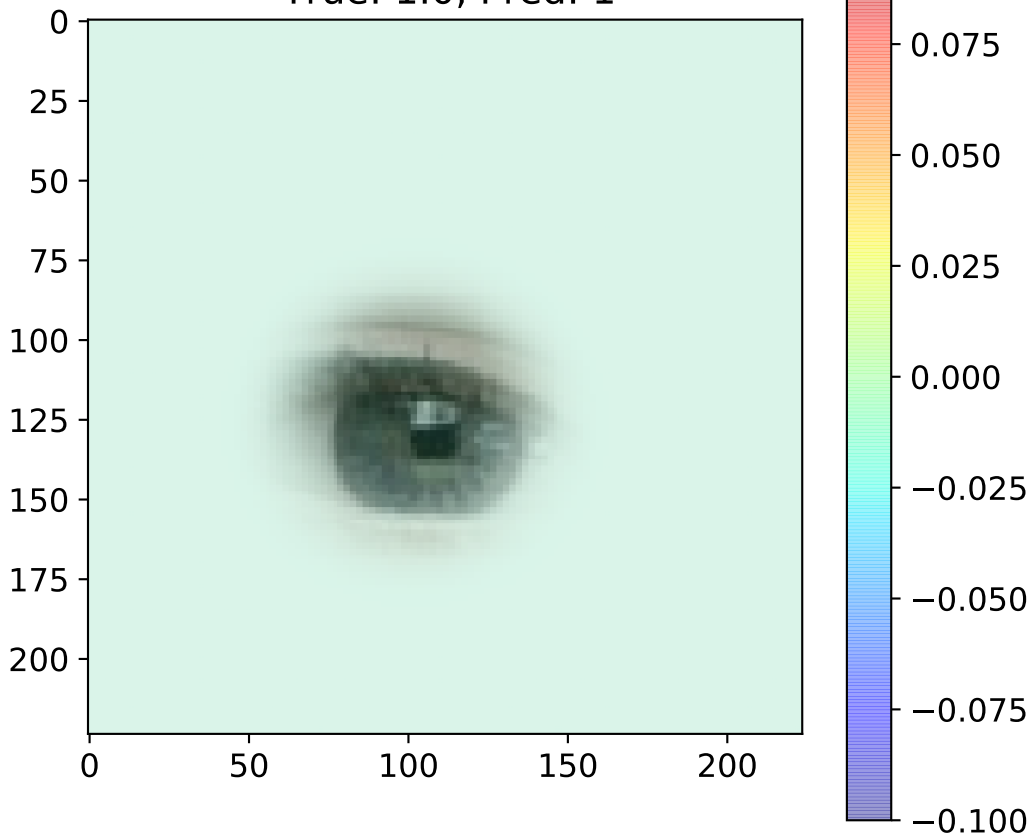
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



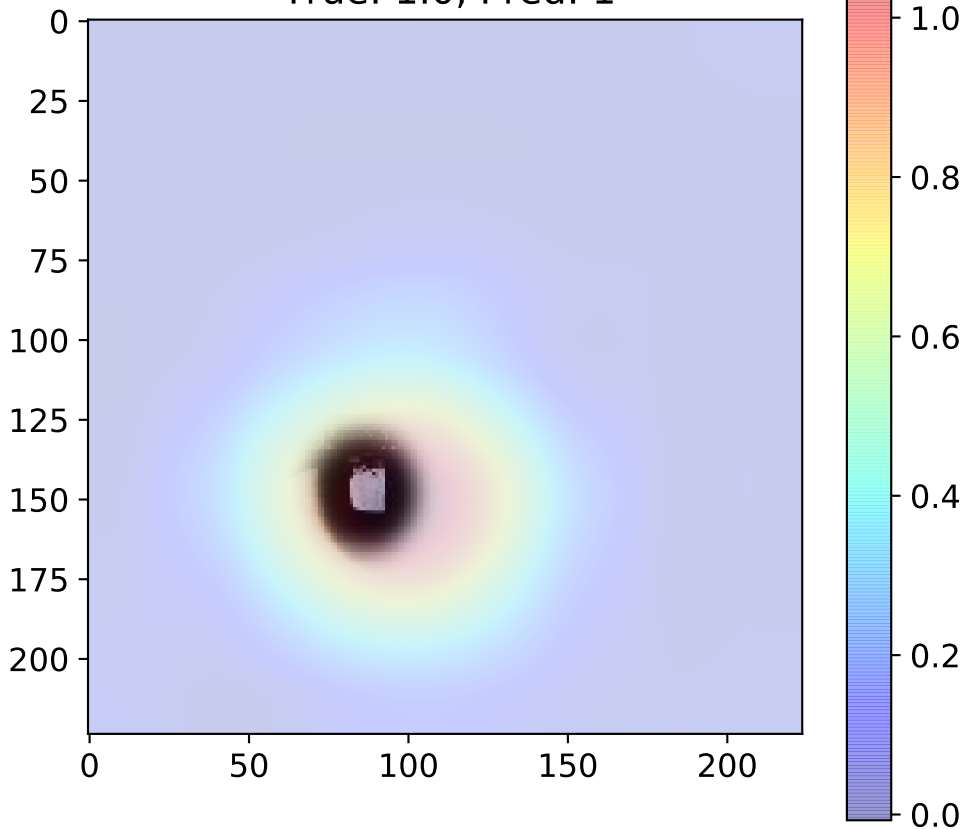
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



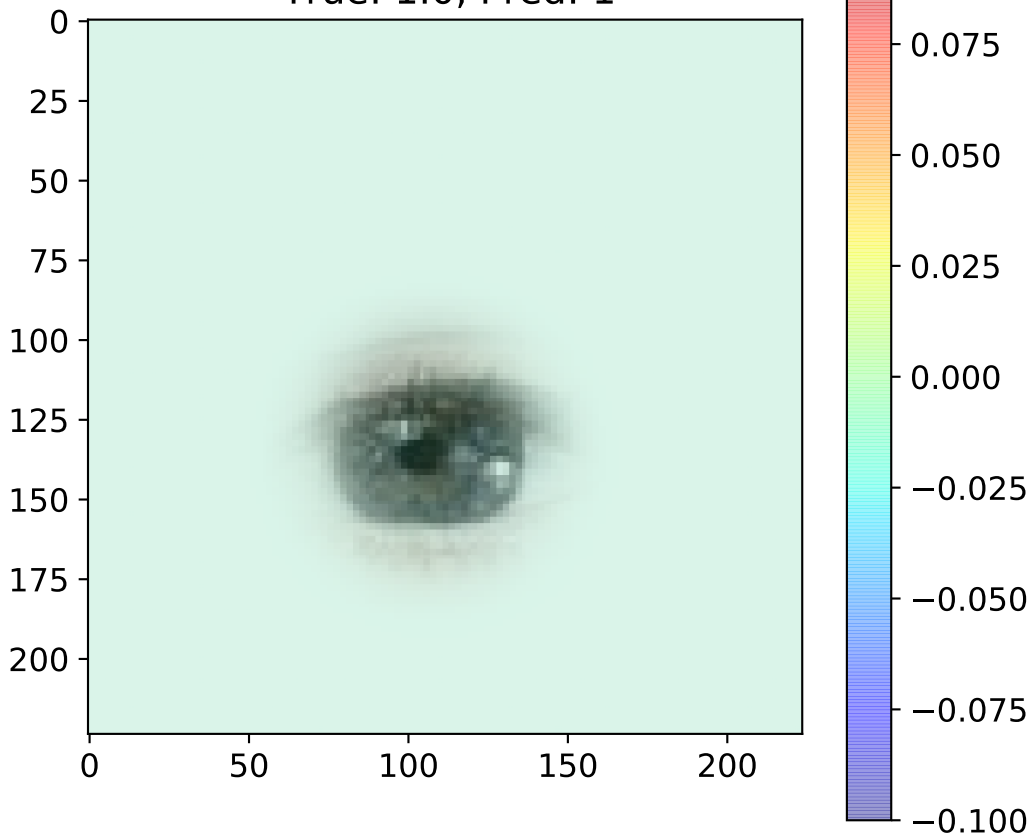
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



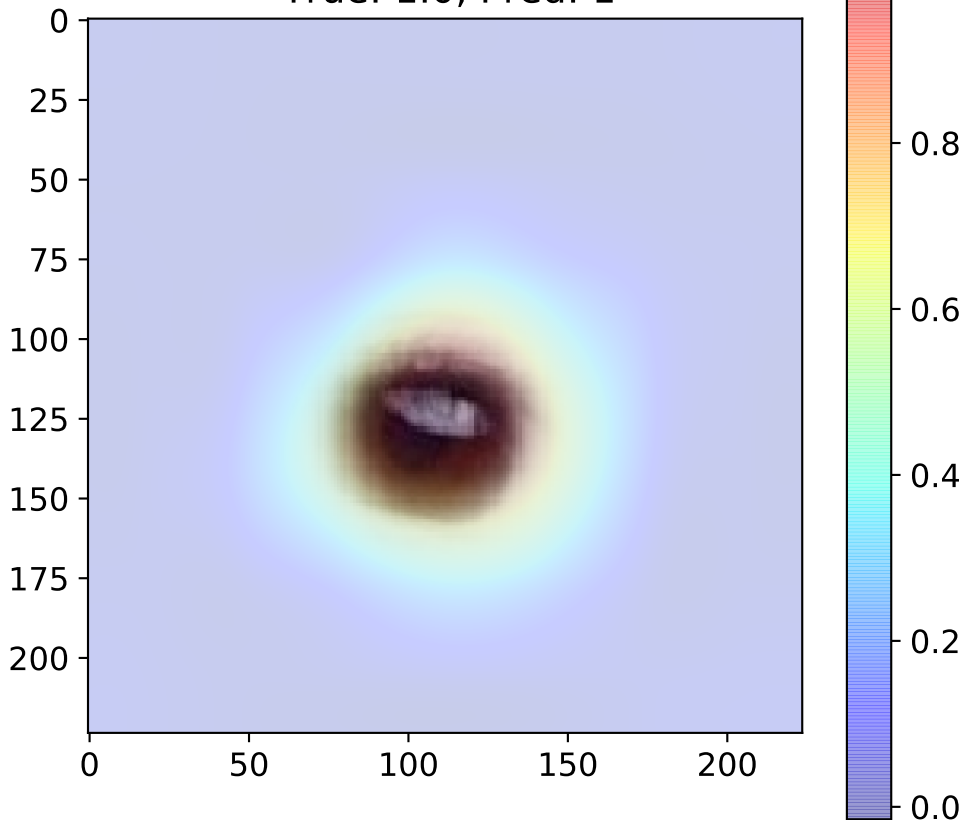
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



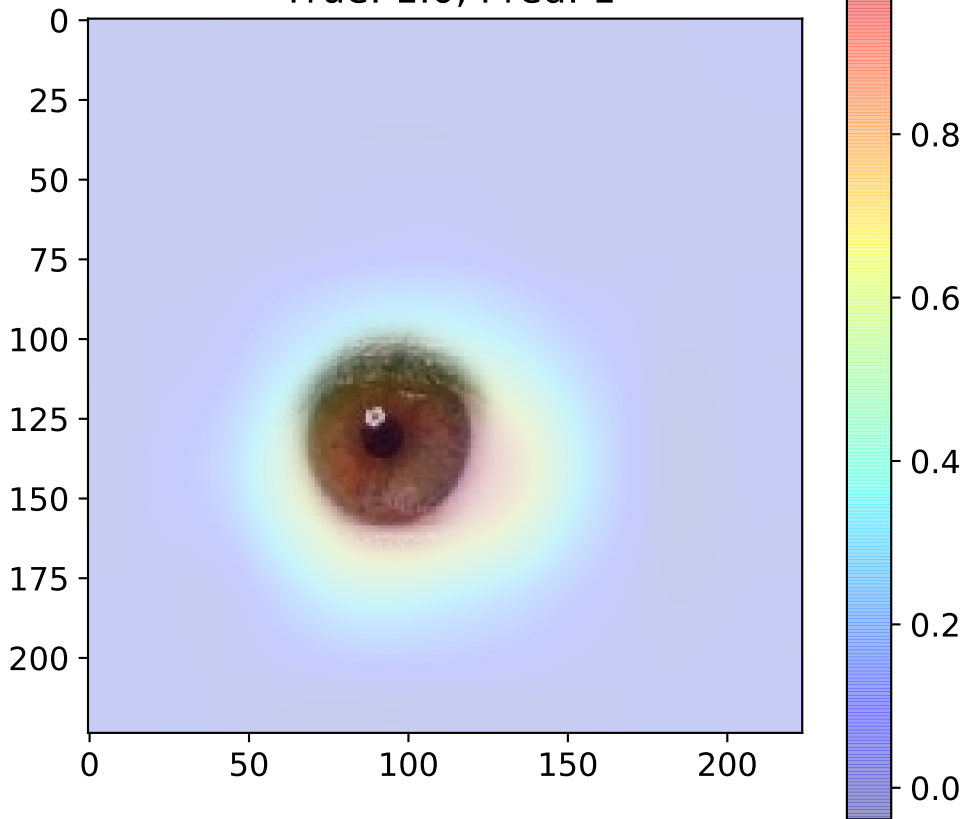
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



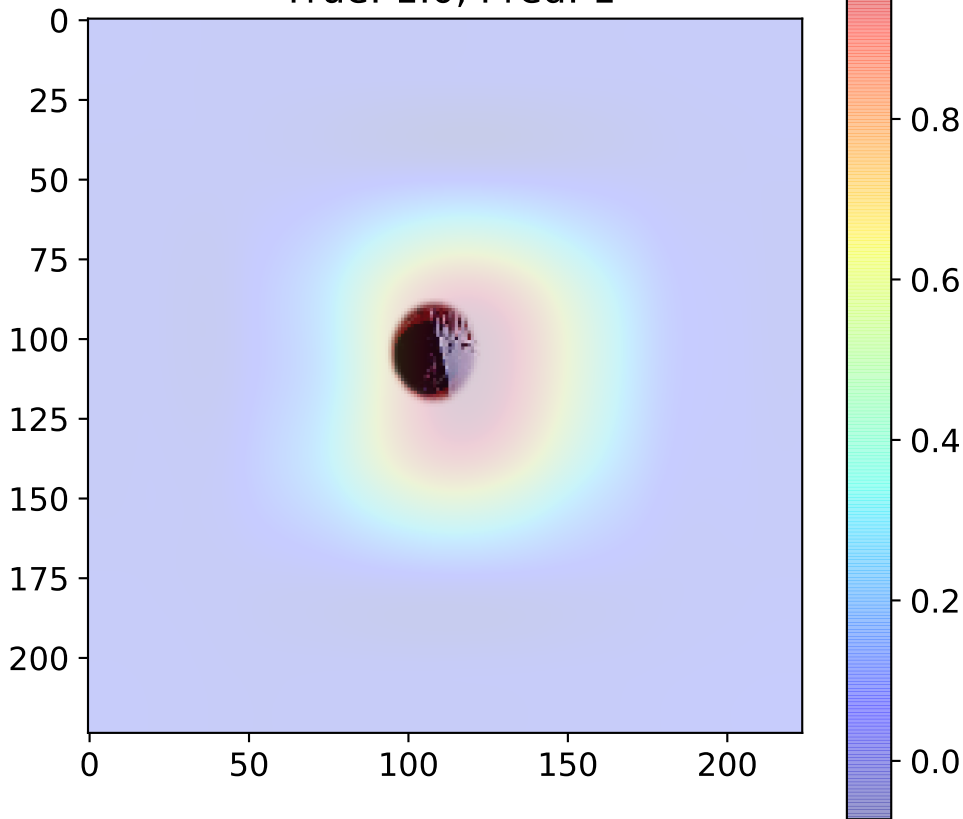
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



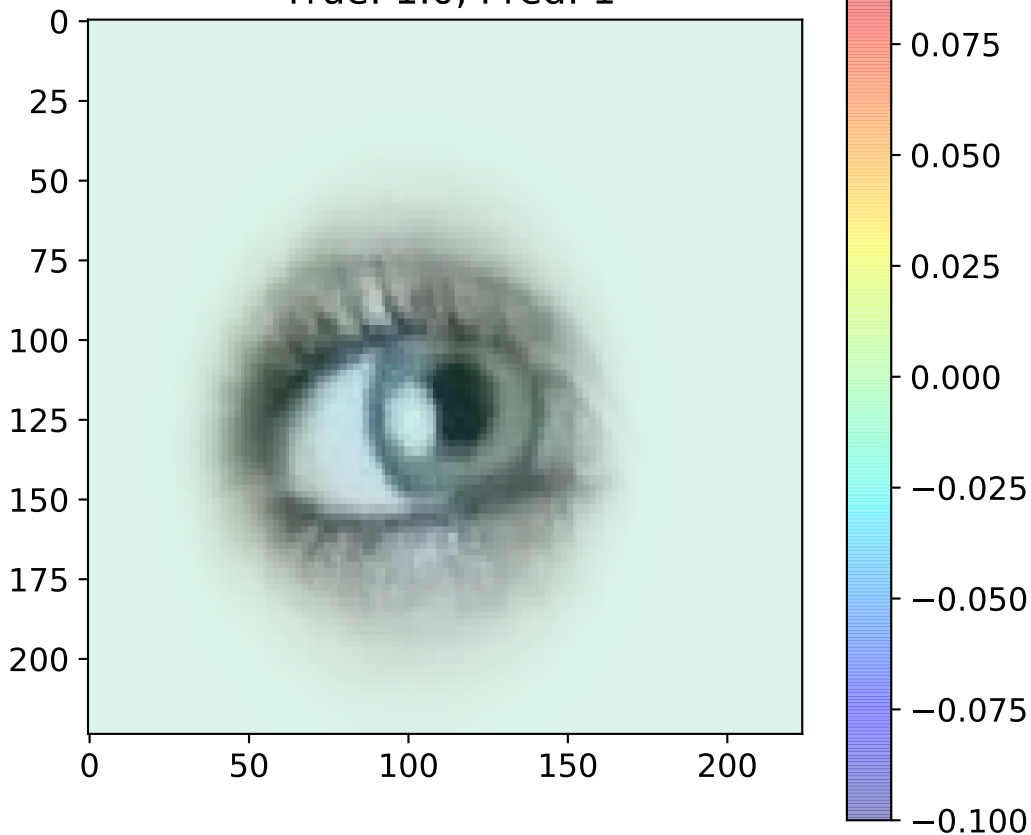
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



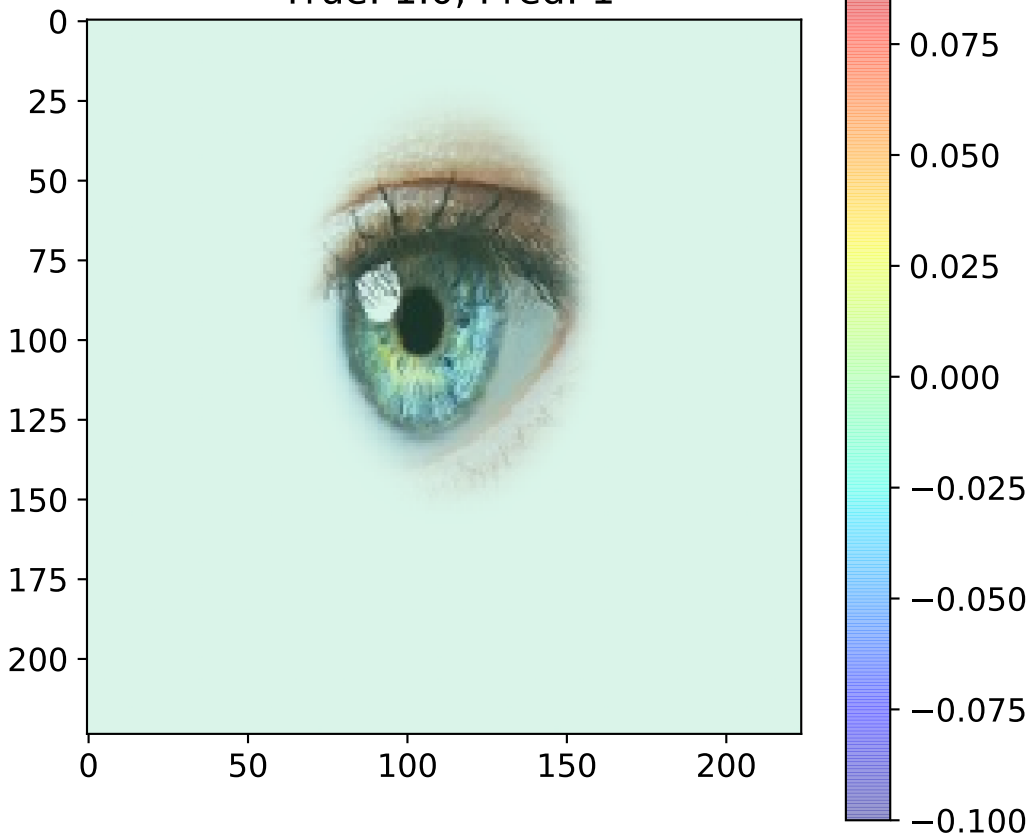
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



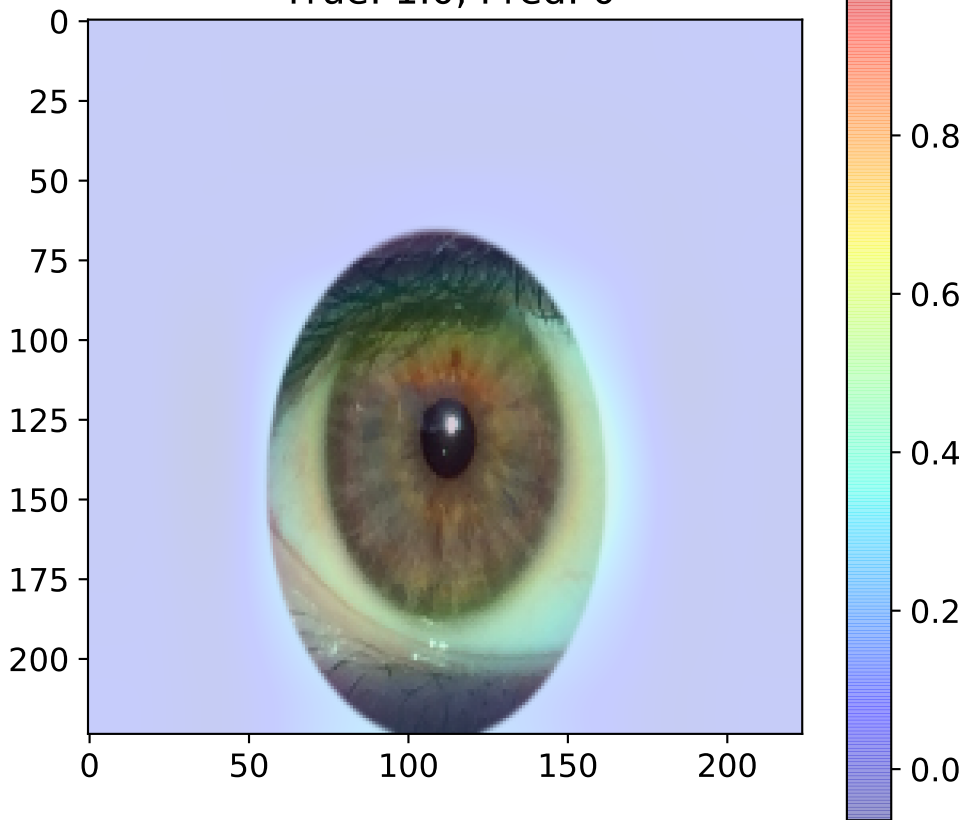
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



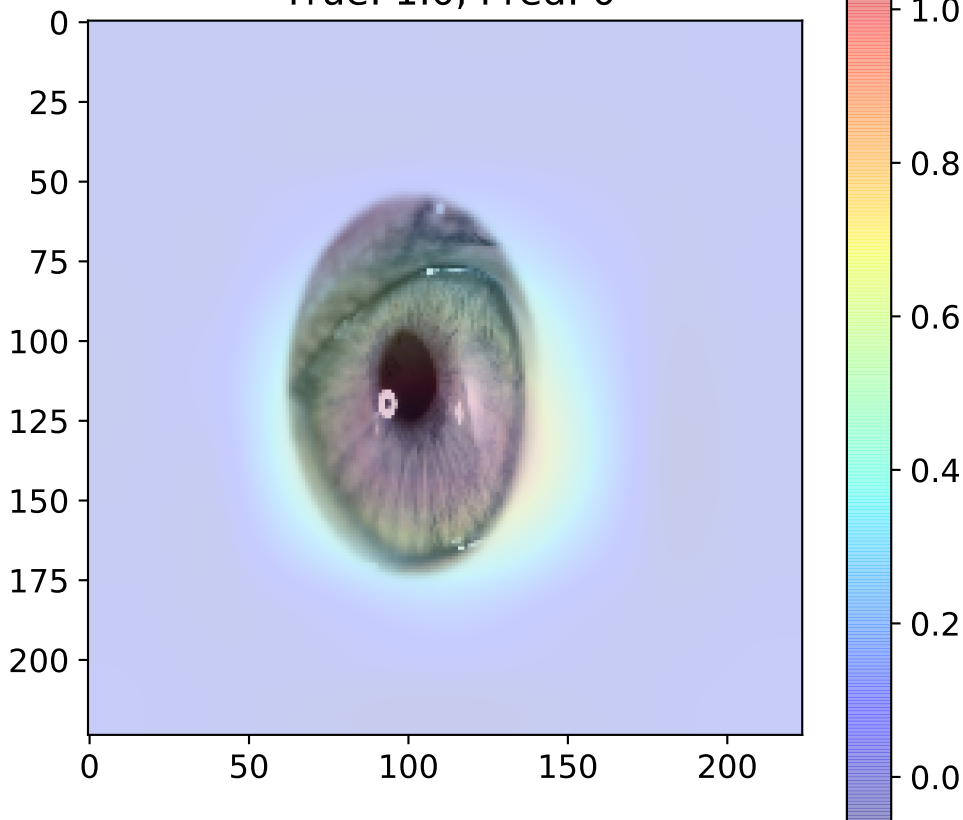
Correctly Classified (Best Fold)
True: 1.0, Pred: 1



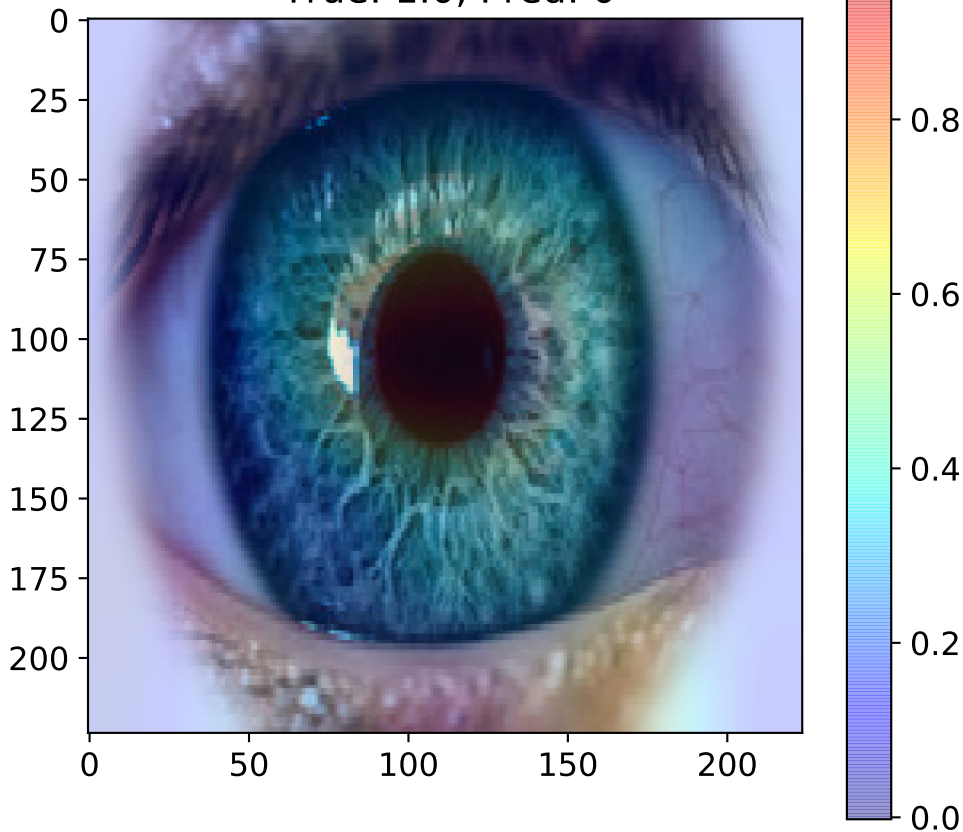
Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0



Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0

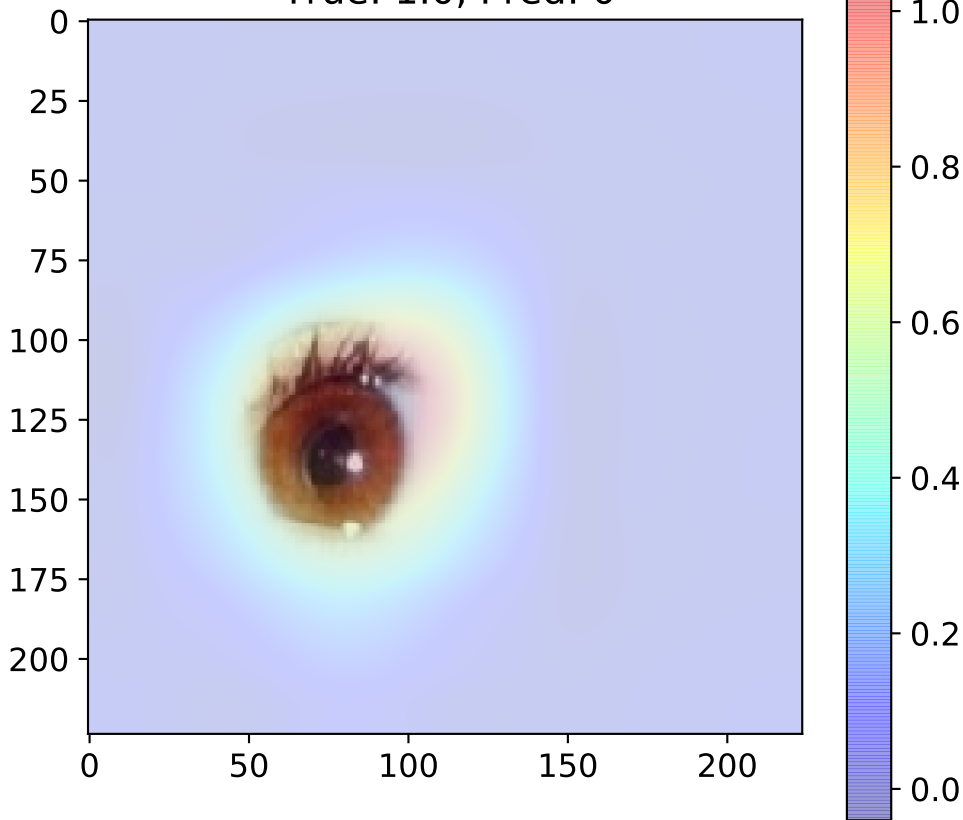


Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0

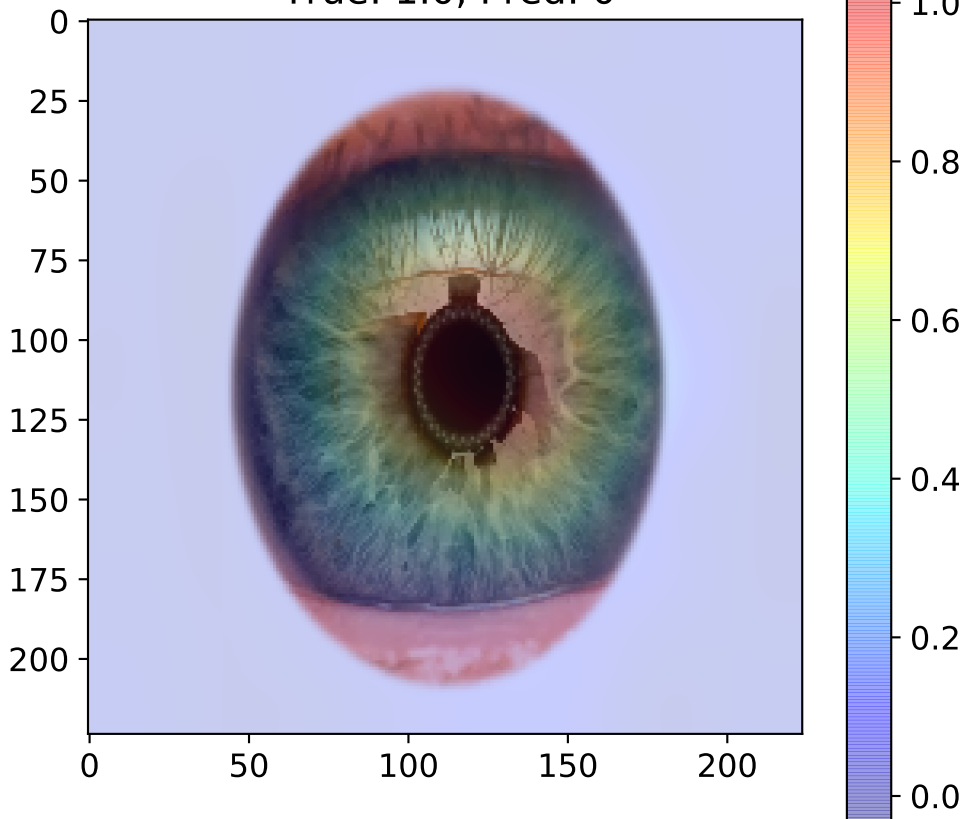


Incorrectly Classified (Best Fold)

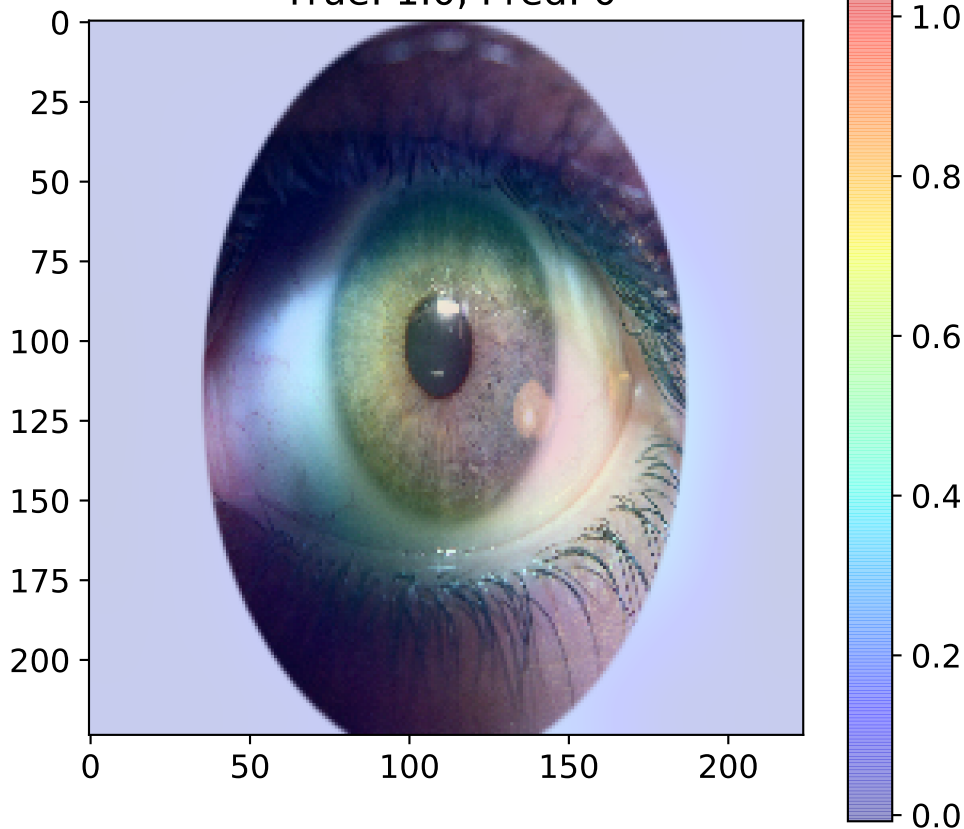
True: 1.0, Pred: 0



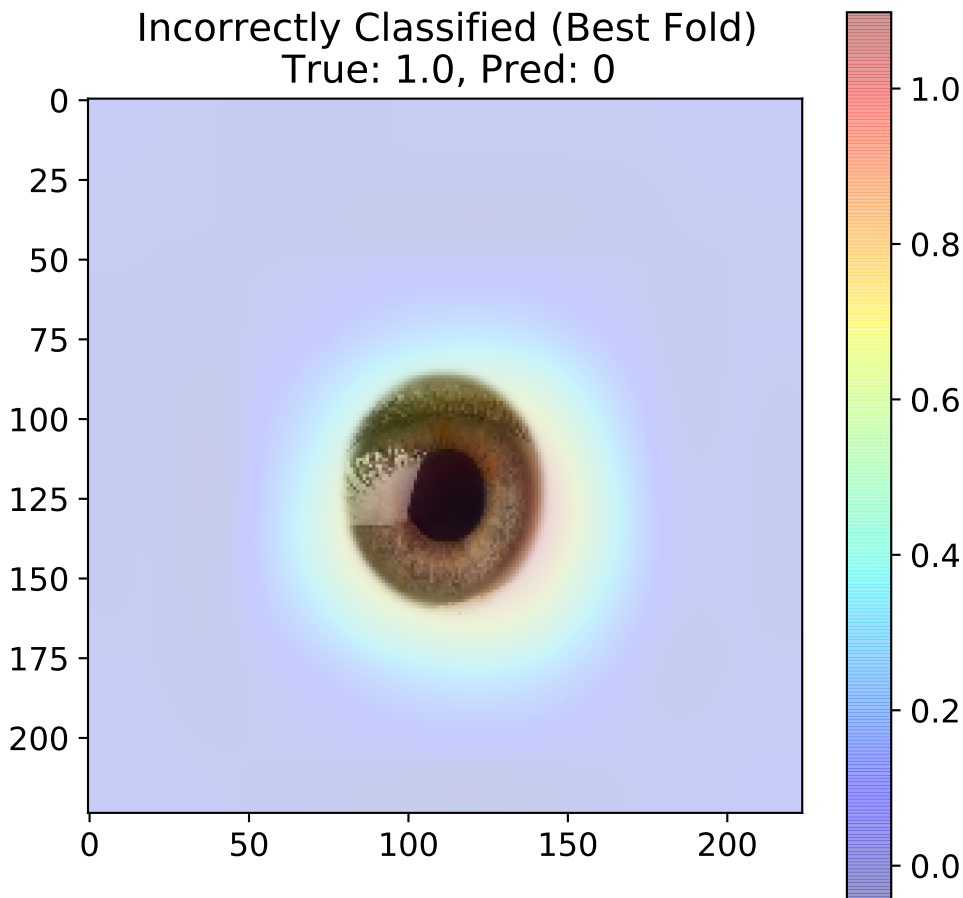
Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0



Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0



Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0



Incorrectly Classified (Best Fold)
True: 1.0, Pred: 0

